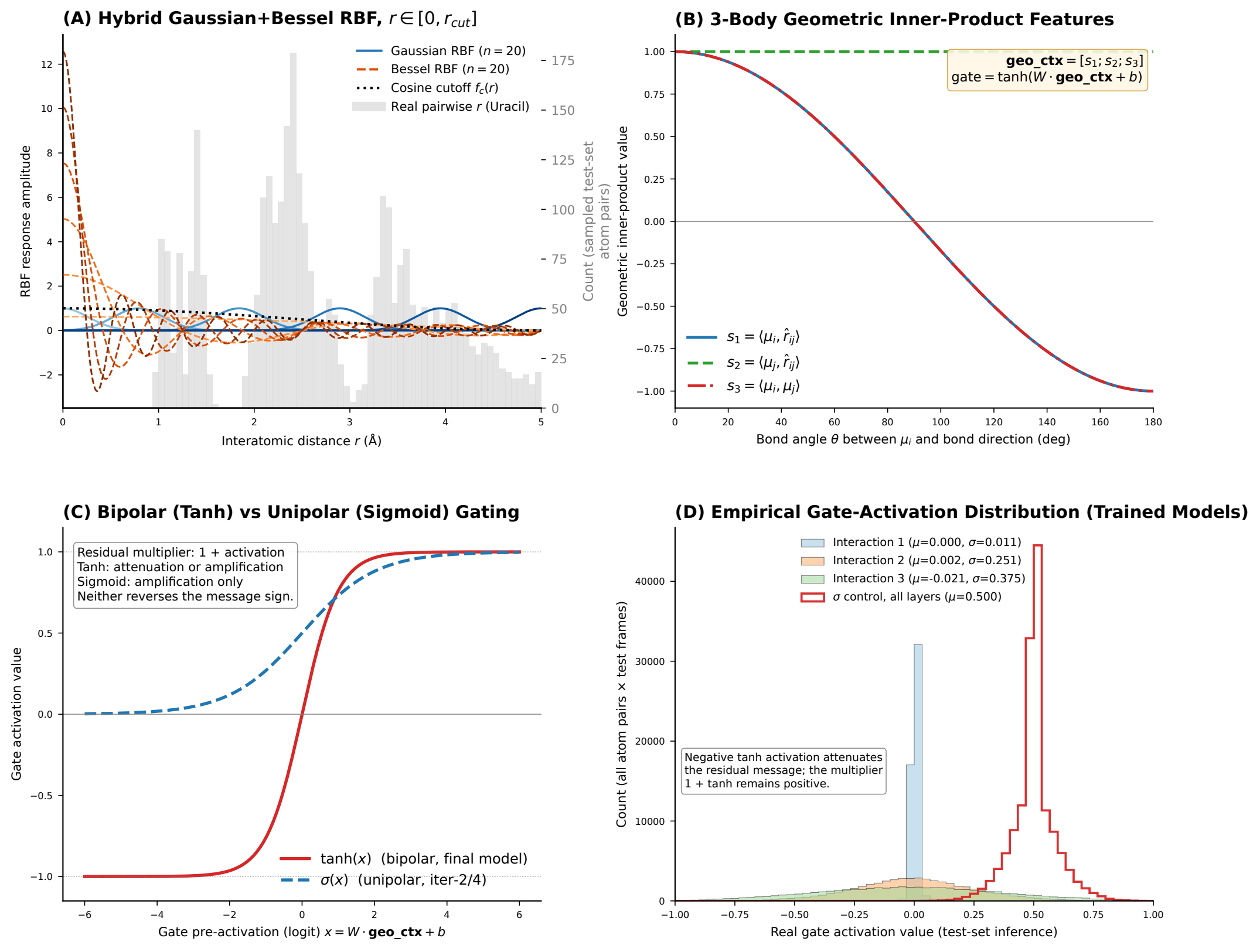

(A) Hybrid Gaussian+Bessel RBF, $r \in [0, r_{cut}]$
Gaussian RBF ($n = 20$)
Bessel RBF ($n = 20$)
Cosine cutoff $f_c(r)$
Real pairwise r (Uracil)
RBF response amplitude
Count (sampled test-set atom pairs)
Interatomic distance r (Å)
(B) 3-Body Geometric Inner-Product Features
$\mathbf{geo_ctx} = [s_1; s_2; s_3]$
$\text{gate} = \tanh(W \cdot \mathbf{geo_ctx} + b)$
$s_1 = \langle \mu_i, \hat{r}_{ij} \rangle$
$s_2 = \langle \mu_j, \hat{r}_{ij} \rangle$
$s_3 = \langle \mu_i, \mu_j \rangle$
Geometric inner-product value
Bond angle θ between μ_i and bond direction (deg)
(C) Bipolar (Tanh) vs Unipolar (Sigmoid) Gating
Residual multiplier: 1 + activation
Tanh: attenuation or amplification
Sigmoid: amplification only
Neither reverses the message sign.
tanh(x) (bipolar, final model)
$\sigma(x)$ (unipolar, iter-2/4)
Gate activation value
Gate pre-activation (logit) $x = W \cdot \mathbf{geo_ctx} + b$
(D) Empirical Gate-Activation Distribution (Trained Models)
Interaction 1 (μ=0.000, σ=0.011)
Interaction 2 (μ=0.002, σ=0.251)
Interaction 3 (μ=-0.021, σ=0.375)
σ control, all layers (μ=0.500)
Negative tanh activation attenuates the residual message; the multiplier 1 + tanh remains positive.
Count (all atom pairs × test frames)
Real gate activation value (test-set inference)